

A Link Between Summation, Differentiation, and Laguerre Polynomials

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The following is a delightful theorem which constitutes an intriguing connection between the act of evaluating a function at distinct points and taking its derivative. (Eqn. 1)

$$\boxed{-\sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=0}^n (-1)^k \binom{n}{k} f(z + ak) = af'(z)}$$

(1)

I consider this process a converse to the function's Taylor-series expansion which, in essence, performs the reverse:

$$f(z + a) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a) z^n}{n!}$$

Additionally, it is evident that the inner summand constitutes a repeated forward difference operation, quite resemblant to a newton series ($\Delta_a[f(x)] = f(x + a) - f(x)$);

$$-\sum_{n=1}^{\infty} \frac{(-1)^n \Delta_a^n[f(z)]}{n} = af'(z)$$

In any case, equation (1) provides a useful link between functions well established over the integers to their potentially trickier derivatives. The following limiting rearrangement exists for the interchange of summation order; (Nth Harmonic Number = H_N)

$$\begin{aligned} \lim_{N \rightarrow \infty} -\sum_{n=1}^N \left(\sum_{k=1}^n \frac{(-1)^k}{n} \text{nCr}(n, k) f(z + ak) + \frac{1}{n} f(z) \right) \\ \lim_{N \rightarrow \infty} -\sum_{k=1}^N \frac{(-1)^k}{k} \text{nCr}(N, k) f(z + ak) - f(z) H_N \end{aligned}$$

1 Proof

Assuming f is analytic and that equation (1) converges, we gather the immediate;

$$-\sum_{m=0}^{\infty} \frac{f^{(m)}(x)}{m!} a^m \sum_{n=1}^N \frac{1}{n} \sum_{k=0}^n (-1)^k \text{nCr}(n, k) k^m$$

Noting the setup of the binomial theorem, construct;

$$\eta_n(x) = \sum_{k=0}^n (-1)^k \text{nCr}(n, k) e^{kx} = (1 - e^x)^n$$

Which, through differentiation, can be substituted as $\sum_{k=0}^n (-1)^k \text{nCr}(n, k) k^m = \frac{d^m}{dx^m} \eta_n(x)_{\{x=0\}}$

$$- \sum_{m=0}^{\infty} \frac{f^m(z)}{m!} a^m \frac{d^m}{dx^m} \left(\sum_{n=1}^N \frac{1}{n} \eta_n(x) \right)_{\{x=0\}}$$

$$- \sum_{m=0}^{\infty} \frac{f^m(z)}{m!} a^m \frac{d^m}{dx^m} (-\ln(1 - (1 - e^x)))_{\{x=0\}} \Rightarrow \sum_{m=0}^{\infty} \frac{f^m(z)}{m!} a^m \frac{d^m}{dx^m} x_{\{x=0\}}$$

Which in turn extracts only the 1st derivative;

$$\frac{f'(z) a^1}{1} \cdot 1$$

For the enthusiasts of operational calculus, another proof could merely rely on the established relation of $e^D = T$ between the derivative operator D and shift operator T ;

$$- \sum_{n=1}^{\infty} \sum_{k=0}^n \frac{1}{n} (-1)^k \text{nCr}(n, k) f(ak + z) \Rightarrow \left[- \sum_{n=1}^{\infty} \sum_{k=0}^n \frac{1}{n} (-1)^k \text{nCr}(n, k) T^{ak} \right] f(z)$$

$$\left[- \sum_{n=1}^{\infty} \frac{1}{n} (1 - e^{Da})^n \right] f(z) \Rightarrow [\ln(1 - (1 - e^{Da}))] f(z) \Rightarrow af'(z)$$

2 Example: Euler-Mascheroni

Since factorials are well incorporated into sums, one can utilize the identity $\frac{d}{dx} \ln(x!)_{\{x=0\}} = \frac{d}{dx} \frac{1}{x!}_{\{x=0\}} = -\gamma$ to produce various series:

On the one hand by using $1/k!$, we have a splendid sum over the Laguerre Polynomials $L_n(x) = \sum_{k=0}^n \text{nCr}(n, k) \frac{(-1)^k}{k!} x^k$

$$\sum_{n=1}^{\infty} \frac{L_n(1)}{n} = -\gamma \quad (2)$$

And on the other hand the following identity can be developed;

$$\lim_{N \rightarrow \infty} \sum_{k=1}^N (-1)^k \ln(k!) \sum_{n=k}^N \text{nCr}(n, k) \frac{1}{n} \Rightarrow \lim_{N \rightarrow \infty} \sum_{k=1}^N \ln(k!) \text{nCr}(N, k) \frac{(-1)^k}{k} = \gamma$$

Taking inspiration from the product identity $e^\gamma = \prod_{n=1}^{\infty} \prod_{k=0}^n (k+1)^{\frac{(-1)^{k+1}}{n+1} \text{nCr}(n, k)}$, we can write;

$$\lim_{N \rightarrow \infty} \frac{1}{N!} \ln \left(\prod_{k=1}^N (k!)^{\text{nCr}(N, k) N!^{\frac{(-1)^k}{k}}} \right) = \gamma \quad e^\gamma = \prod_{n=1}^{\infty} \prod_{k=0}^n k!^{(-1)^k \text{nCr}(n, k) \frac{1}{n}} \quad (3)$$

Or finally, for reasonably fast convergence;

$$\gamma = \lim_{N \rightarrow \infty} \text{and/or} \lim_{a \rightarrow \infty} - \sum_{k=1}^N \frac{\text{nCr}(N, k)}{k} (-1)^k \frac{a}{(\frac{k}{a})!} - aH_N$$

$$- \sum_{k=0}^{\text{floor}(\frac{N}{a})} \sum_{m=1}^a \frac{\text{nCr}(N, ak+m)}{ak+m} (-1)^{(ak+m)} \frac{a}{(k + \frac{m}{a})!} - aH_N \quad (4)$$

Fixing a may be the only practical computational method as fractional arguments of the Gamma function soon appear, though $\Gamma(x + 1/q)$ identities as falling products do exist.

$a = 2$

$$\lim_{N \rightarrow \infty} -2 \sum_{n=1}^N \frac{1}{n} \sum_{k=0}^{\text{floor}(\frac{n}{2})} \left(\text{nCr}(n, 2k) \frac{1}{k!} - \frac{4}{\sqrt{\pi}} \text{nCr}(n, 2k+1) \frac{4^k (k+1)!}{(2k+2)!} \right)$$

$a = 4$

$$\lim_{N \rightarrow \infty} -4H_N - \text{nCr}(N, 4) - \sum_{k=0}^{\text{floor}(\frac{N}{4})} 4^{(k+2)} \text{nCr}(N, 4k+1) a_{N,k}$$

$$a_{N,k} = \frac{-1}{4k+1} \frac{1}{\Gamma(\frac{1}{4}) \prod_{j=1}^{k+1} (4j-3)} - \frac{(N-4k-2)(N-4k-1)}{(4k+3)^2 (2k+1)} \frac{\frac{\Gamma(\frac{1}{4})}{2\sqrt{2\pi}}}{\prod_{j=1}^{k+1} (4j-1)} + \frac{N-4k-1}{(2k+1)^2} \frac{(k+1)!}{4\sqrt{\pi} (2k+2)!}$$

3 Miscellaneous

The zeta function is yet another function well-defined over specific integers. The following is an immediate result from the even-integer Bernoulli relation, providing a nearly-Newton series for $\zeta'(2p)$ (for integer p).

$$\frac{(-1)^p}{4} \sum_{n=1}^{\infty} \sum_{m=p}^{n+p} \frac{1}{n} \text{nCr}(n, m-p) \frac{(2\pi)^{2m}}{(2m)!} B_{2m} = \zeta'(2p)$$

For functions with well-defined integrals over the integers, one can iterate the integral inside of the running summand. Here, $\int_a^{a+1} \ln(\Gamma(t)) dt = a \ln(a) - a + \frac{1}{2} \ln(2\pi)$ is used as an example.

$$\begin{aligned} -\ln \Gamma(q) &= \sum_{n=1}^{\infty} \sum_{k=0}^n \frac{1}{n} (-1)^k \text{nCr}(n, k) \left(\sum_{m=1}^{k+q-1} m \ln(m) - \frac{(k+q-1)(k+q)}{2} + \frac{1}{2} \ln(2\pi)(k+q) \right) \\ &= \lim_{N \rightarrow \infty} \sum_{k=1}^N \frac{(-1)^k}{k} \text{nCr}(N, k) \ln(H(k+q)) + \ln(H(q)) H_N + \ln(q) + \frac{1}{2} + q - \frac{1}{2} \ln(2\pi) \end{aligned} \quad (5)$$

For integer q and for the Hyperfactorial function $H(n) = \prod_{i=1}^n i^i$.

Finally, it is intriguing to see also the failure of the formula for where the behavior of a function at integers is deceptive of its general behavior. $\sin(\pi x)$ for instance, evaluates always to 0 when summed over the integers, but has non-vanishing derivatives at many points (assume a, z are integers). Identical shortcomings appear in the function's Newton Series, moreover.

$$-\sum_{n=1}^N \sum_{k=0}^n \frac{1}{n} (-1)^k \text{nCr}(n, k) \sin(\pi a k + \pi z) = 0 \neq a \frac{d}{dx} \sin(\pi x)_{\{x=z\}}$$

...And these examples are merely a few of the endless possibilities of applications between summation and differentiation!

4 Laguerre-Style Taylor Series

On the subject of Laguerre polynomials $L_n(x) = \sum_{k=0}^n \text{nCr}(n, k) \frac{(-1)^k}{k!} x^k$ we have the following Laguerre 'rendition' of a Taylor polynomial which has stikingly similar form to Equation (1).

$$\sum_{n=0}^{\infty} t^{(n+1)} \sum_{k=0}^n \text{nCr}(n, k) \frac{(-1)^k}{k!} \frac{f^{(k+1)}(z)}{k+1} = f(z) - f\left(z - \frac{t}{1-t}\right) \quad (6)$$

Using a subtly modified Laguerre polynomial as seen in (6) and setting $t = 1/2$, perhaps this can be specialized to: (for backwards difference operator ∇)

$$\frac{1}{2} \sum_{n=0}^{\infty} \frac{L_n^*(D)}{2^n} = \nabla$$

As with before, binomial coefficient algebra can be used to derive the expression:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k!} \frac{f^{(k+1)}(z)}{k+1} t^{(k+1)} \sum_{n=0}^{\infty} \text{nCr}(n+k, n) t^n + t f'(z) \sum_{n=0}^{\infty} t^n$$

$$\begin{aligned}
& - \sum_{k=2}^{\infty} \frac{(-1)^{(k)}}{(k)!} f^{(k)}(z) \left(\frac{t}{1-t} \right)^{(k)} + f'(z) \frac{t}{1-t} \\
& - \sum_{k=0}^{\infty} \frac{(-1)^{(k)}}{(k)!} f^{(k)}(z) \left(\frac{t}{1-t} \right)^{(k)} + \left(\frac{(-1)^{(0)}}{(0)!} f^{(0)}(z) \left(\frac{t}{1-t} \right)^{(0)} + \frac{(-1)^{(1)}}{(1)!} f^{(1)}(z) \left(\frac{t}{1-t} \right)^{(1)} \right) + f'(z) \frac{t}{1-t} \\
& = f(z) - f\left(z - \frac{t}{1-t}\right)
\end{aligned}$$

Or, under the formal use of linear operators $e^D = T$ and the Laguerre polynomial generating function;

$$\begin{aligned}
& \left[\sum_{n=0}^{\infty} t^{(n+1)} \sum_{k=0}^n \text{nCr}(n, k) \frac{(-1)^k}{k!} \frac{D^{(k+1)}}{k+1} \right] f(z) = \left[\sum_{n=0}^{\infty} t^{(n+1)} \int_0^D L_n(x) dx \right] f(z) \\
& = t \int_0^D \left[\sum_{n=0}^{\infty} t^n L_n(x) \right] dx f(z) = t \left[\int_0^D \frac{e^{-\frac{tx}{1-t}}}{1-t} dx \right] f(z) = \left[1 - (e^D)^{\frac{-t}{1-t}} \right] f(z) = f(z) - f\left(z - \frac{t}{1-t}\right)
\end{aligned}$$